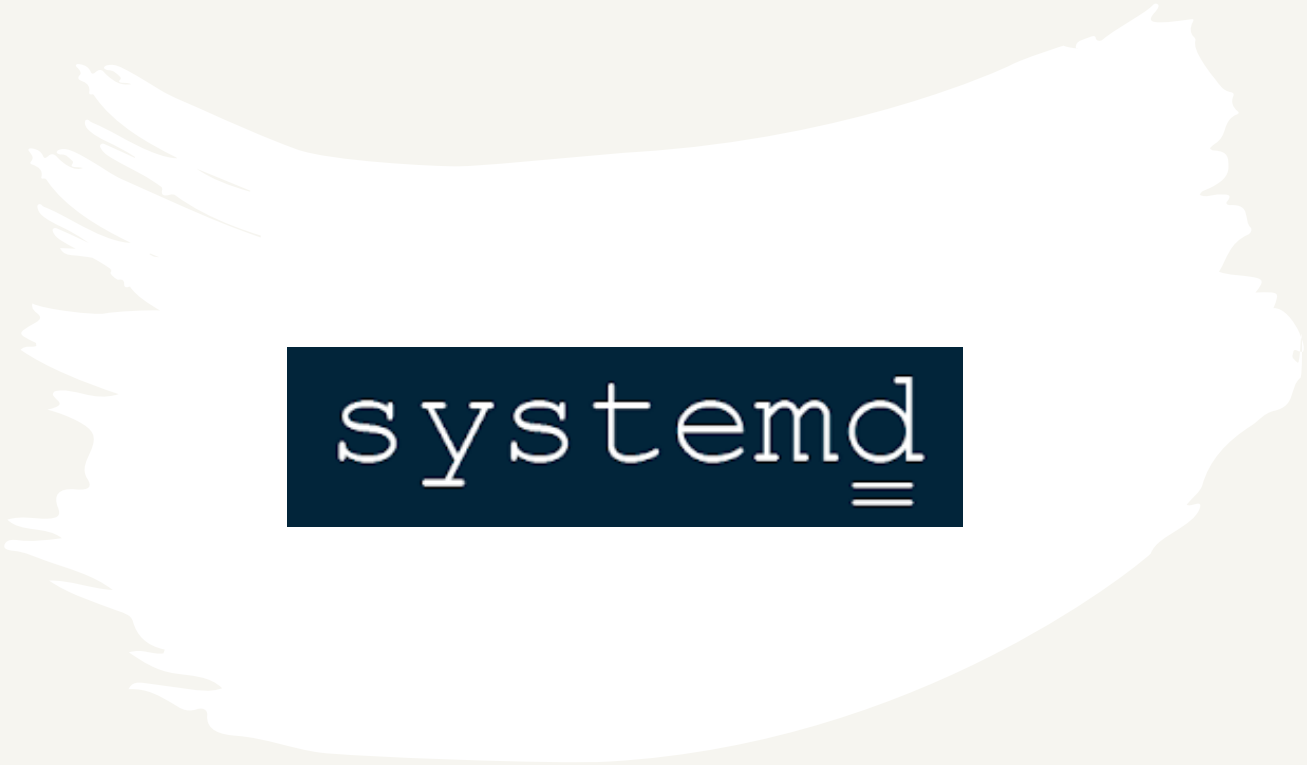


The logo for systemd, featuring the word "systemd" in a white, lowercase, monospaced font. The letters 'y' and 'd' have horizontal lines extending from their bases. The text is centered within a dark blue rectangular box, which is itself centered within a larger, white, irregular shape that resembles a torn piece of paper or a brushstroke. The entire graphic is set against a light beige background.

systemd

systemd + IoT Devices

Making your Raspberry Pi behave like a real device

What is systemd?

Default Linux
init system

Manages
services and
boot

Ensures
reliability

Why systemd?



AUTO START ON BOOT



RESTART ON FAILURE



RUNS IN BACKGROUND

IoT Context

Devices must
run
unattended

Must recover
from crashes

systemd
enables this

Creating a systemd Service

Turn a Python script into a reliable background service

1. Test app manually (python sense_api.py)
2. Create service file in /etc/systemd/system/
3. Configure: User, WorkingDirectory, ExecStart, Restart(see Lab)
4. Reload: systemctl daemon-reload
5. Start & check: systemctl start/status
6. Enable on boot: systemctl enable

Common Issues:

- Wrong ExecStart path
- Not using virtual environment
- Forgot daemon-reload

Final Result: Always running, auto-starting, self-recovering service

Commands

- `systemctl start/stop/status`
- `systemctl enable`
- `journalctl` for logs

IoT System with systemd + mDNS

